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Project Title:	Effects of Hypergravity on Expression of Genes Involved in Regulation of Feeding		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
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COI Name (Institution):	Fuller, Charles A (University of California)		
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Task Description:	The ambient force environment alters regulation of adiposity, energy intake, and energy expenditure. Our long-term goal is to elucidate the molecular mechanisms underlying these alterations. Here we evaluate the effect of 2G on expression of genes involved in the leptin regulatory pathway as well as on other hypothalamic genes. Leptin is secreted by adipocytes in proportion to body fat. It acts centrally in a negative feedback loop to modify food intake. We have previously shown that the stoichiometry between adiposity and circulating leptin differs significantly at 1G vs. microG and vs. 2G. Similarly, the relationship between circulating leptin and food intake is altered by gravity [e.g., after 16 days of 2G exposure, leptin is lower, but food intake is still less than that at 1G]. This project addresses the latter dysfunctional relationship by testing the hypothesis that 2G alters expression of hypothalamic genes whose products can modulate feeding. We will use two approaches. One is to evaluate genes for 6 neuropeptides known to regulate feeding and whose transcription is modulated by leptin and for receptors for five of these neuropeptides. These candidate genes		

	include neuropeptide Y (NPY), agouti-related peptide (AGRP), cocaine-and-amphetamine-regulated transcript (CART), proopiomelanacortin (POMC), orexin, melanin concentrating hormone (MCH), and receptors for NPY, AGRP/a-MSH, MCH, and orexin. NPY, AGRP, MCH, and orexin stimulate feeding; CART and a product of POMC (a-MSH) inhibit feeding. At 1G, low leptin is accompanied by high message levels for NPY, AGRP, orexin, and MCH and low message levels for POMC and CART. Thus, "aberrant" levels of mRNAs at 2 vs. 1G should indicate which, if any, could underlie altered leptin regulation of feeding at 2G. Our second approach of using microarrays for expression analysis of ~1300 hypothalamic genes will provide insight into which other genes exhibit altered expression before and after the rats have acclimated to hypergravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to enhancing our understanding of the effects of prolonged exposure of altered force fields on energy balance and why humans in space do not adequately regulate their food intake when left to their own schedules, the data from this project will increase our knowledge about factors that regulate energy balance (food intake and energy expenditure) at 1G. Such knowledge is critical for enhancing our understanding of how humans regulate body fat storage under conditions of both food scarcity and abundance. The latter is particularly relevant to industrialized nations which are incurring an "epidemic" of obesity with its associated mortalities/morbidities.
Task Progress:	Our long term goal is to identify the mechanisms underlying the altered regulation of adiposity, energy intake, and energy expenditure during exposure to ambient force environments other than 1G. During the most recent period of this project, we have focused on the expression of genes that could underlie the energy balance changes that occur during exposure to hypergravity (2G) via centrifugation. We have previously shown in rodent models that: (1) the feedback relationship between circulating leptin levels and adiposity remains intact during 2G exposure although the stoichiometry differs at 1G vs. 2G vs. microG; and (2) the relationship between circulating leptin and food intake is altered by gravity such that lower concentrations of leptin do not result in an increased food intake adequate to restore body weight or body fat levels. To determine if this alteration involves changes in expression of hypothalamic genes known to be regulated by leptin at 1G, we have used real time PCR to evaluate levels of mRNA. Our data show that exposure to 2G did in fact alter the relationship between leptin and the expression of some neuropeptides. In particular, after 1 week at 2G, orexin expression was significantly lower than that at 1G while expression of other orexigenic neuropeptides were not significantly enhanced. These effects could contribute to blunted signaling for increased feeding. In contrast, the increased expression of anorexic neuropeptides that we observed would be expected to stimulate feeding. Thus, 2G appears to have had a differential effect on the expression of these orexigenic/anorexic neuropeptides which could have contributed to the rat’s decreased food intake. After 8 wks at 2G, when the rats were eating the same amount of food as those at 1G, none of the neuropeptides examined showed significant differences in expression level compared to levels in 1G rats. The fact that adiposity and circulating leptin levels were still significantly less in the 2G vs. 1G rats suggests a new “set point” for the leptin-neuropeptide expression relationship. In addition to candidate neuropeptides, our Affymetrix microarray analysis identified a number of genes that appear to have been affected by the 2G environment (independent of the decrease in food intake). These potential expression changes need to be verified by real time PCR.
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